

JurisCom RegTech: Jurisdiction-Aware Compliance by Design

A config-driven prototype for cross-border regulatory divergence

4 main questions to address:

- Why is this product right for your company?
- What problem does this tool solve that a memo or spreadsheet cannot?
- What regulatory divergence does it handle, and how?
- What does it *not* do — and why did you make that boundary choice?

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The Danger of the Global Compliance Monolith

Focus Question: Why is this product right

Our client: Standard Chartered Bank. It books transactions in 59+ markets, exposing the danger of monolithic compliance engines.

A US-calibrated rule engine creates both missed obligations and alert fatigue.

Missed Obligations

Under-compliance

A high-risk transaction slips through in Singapore because it fails to cross a US dollar threshold.

Legal exposure: the bank misses a mandatory STR obligation.

Alert Fatigue

Over-compliance

A domestic politician triggers massive US-style foreign-PEP reviews unnecessarily.

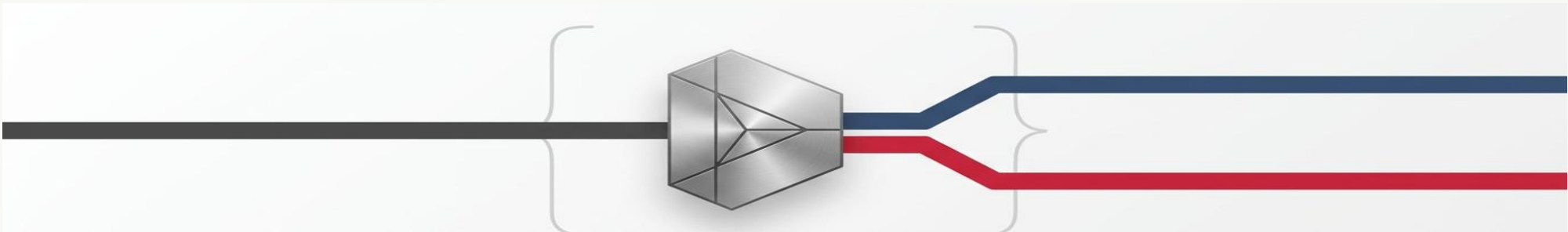
Analyst capacity is consumed by false positives.

A single, inflexible software engine across global operations creates systemic blind spots.

A Tale of Two Regimes: US BSA vs. SG MAS Notice 626

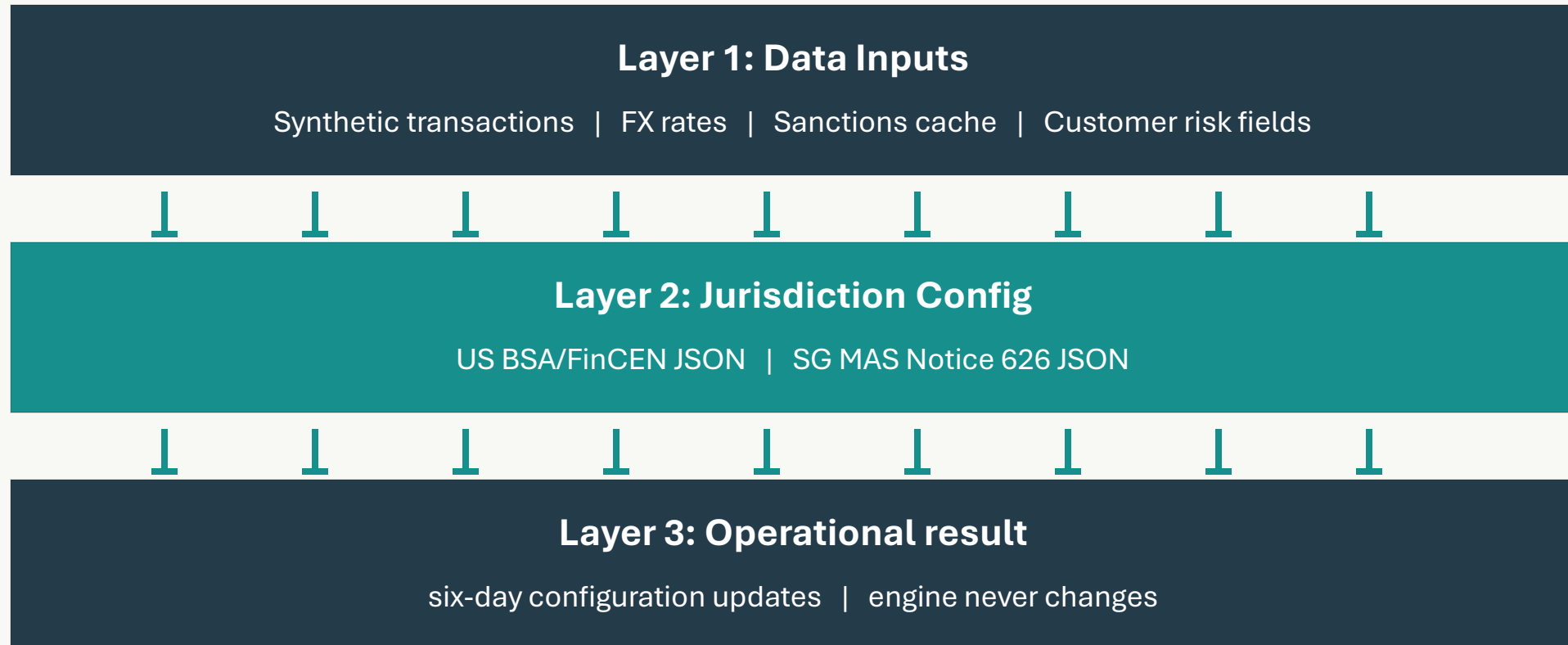
Focus Question: What regulatory divergence does it handle, and how?

Core Dimensions	US Outcomes	SG Outcomes
Cash / One-Off Thresholds	CTR required for cash > USD 10,000.	CDD_ONE_OFF required for non-account holders > SGD 20,000.
Suspicious Reporting	SAR generally requires a USD 5,000 floor.	STR is purely risk-based with no monetary floor.
Politically Exposed Persons (PEPs)	EDD focused heavily on Foreign PEPs.	EDD mandated for both Domestic and Foreign PEPs.
Beneficial Ownership (UBO)	25% bright-line ownership threshold.	25% benchmark + reasonable measures standard.



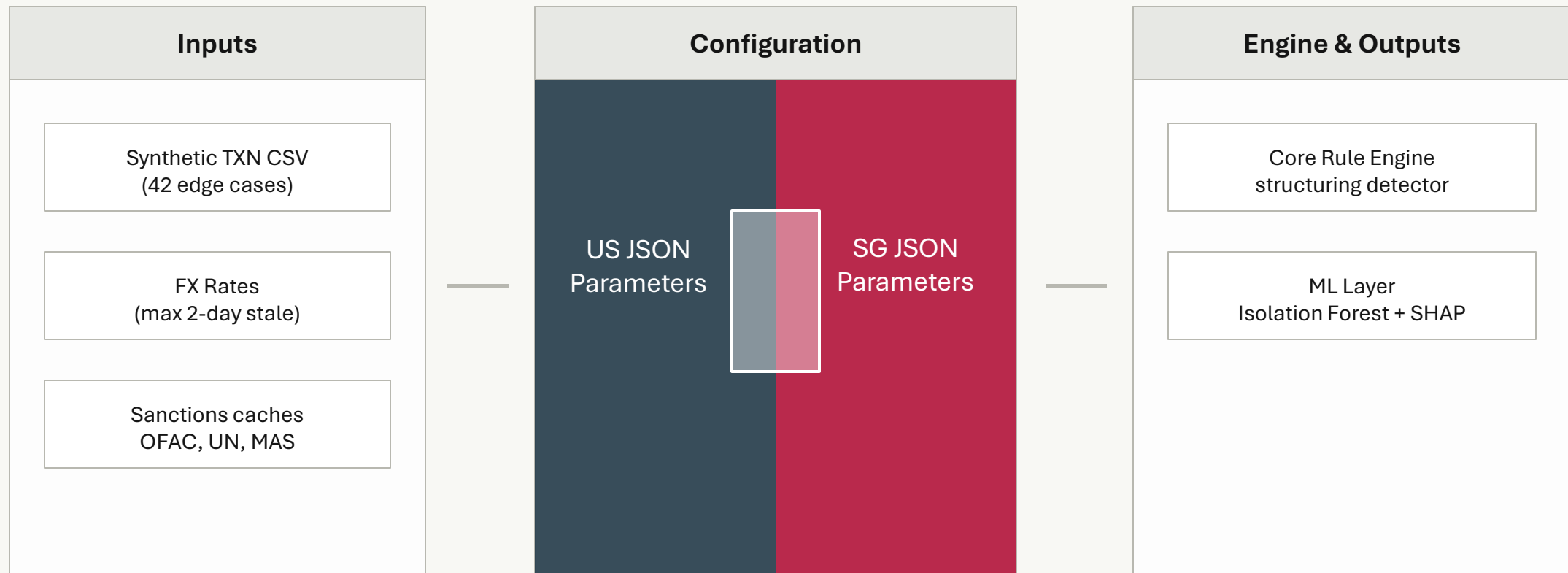
Architecture Innovation: Config Over Code

Focus Question: What does this tool do that a memo or a spreadsheet cannot?



Business value: regulatory logic lives in lightweight JSON files, not hard-coded Python.

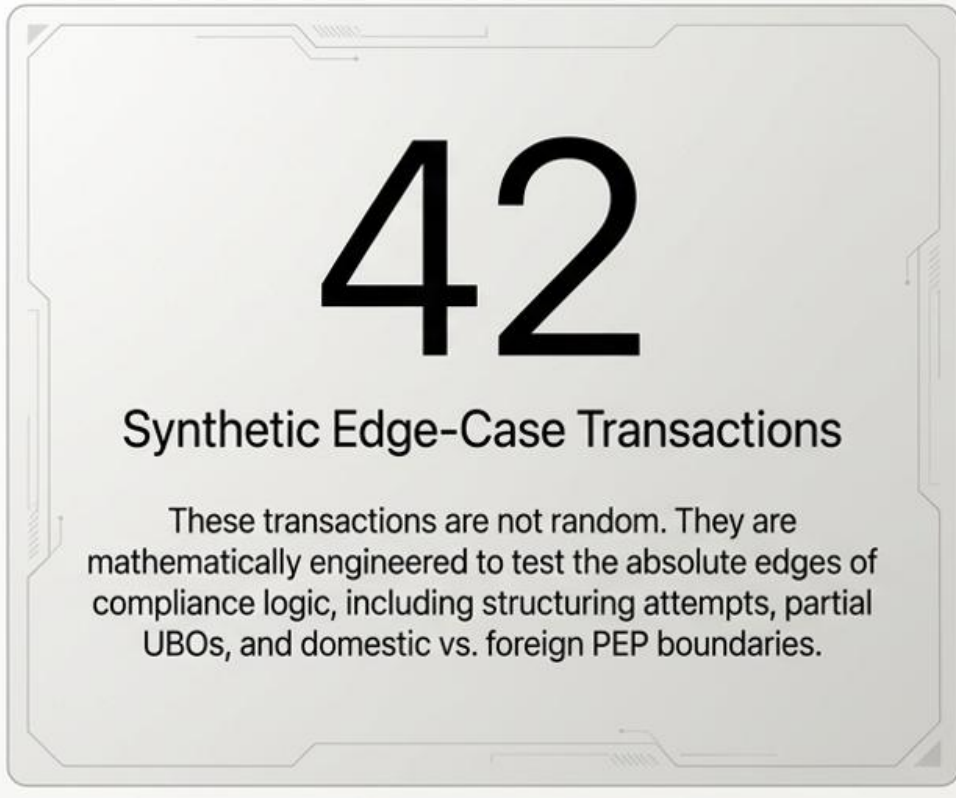
System Architecture: The 3-Layer Stack



Layer 1 and Layer 3 remain stable. Only Layer 2 changes when laws change.

The Input Engine & Alert Taxonomy

Anatomy of the Test Dataset



42

Synthetic Edge-Case Transactions

These transactions are not random. They are mathematically engineered to test the absolute edges of compliance logic, including structuring attempts, partial UBOs, and domestic vs. foreign PEP boundaries.

The 7 Alert Typologies

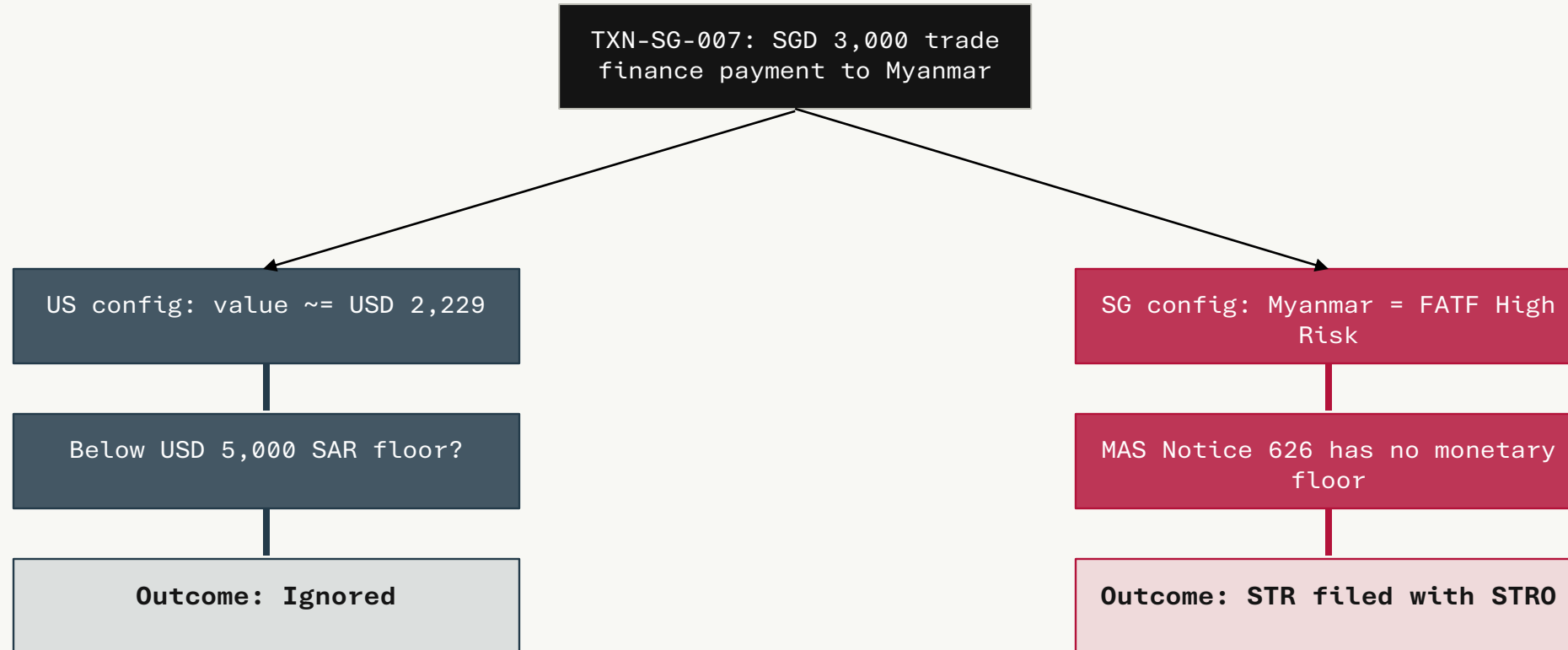
CTR	US cash reporting
CDD_ONE_OFF	SG non-account-holder checks
SAR	US suspicious activity
STR	SG suspicious transaction
SANCTIONS_BLOCK	OFAC, UN, MAS fuzzy matches
EDD_PEP	Enhanced due diligence triggers
BO_FLAG	Beneficial ownership opacity alerts

Proof of Concept: The 4 Key Divergence Cases

Focus Question: What divergence does it handle, and how?

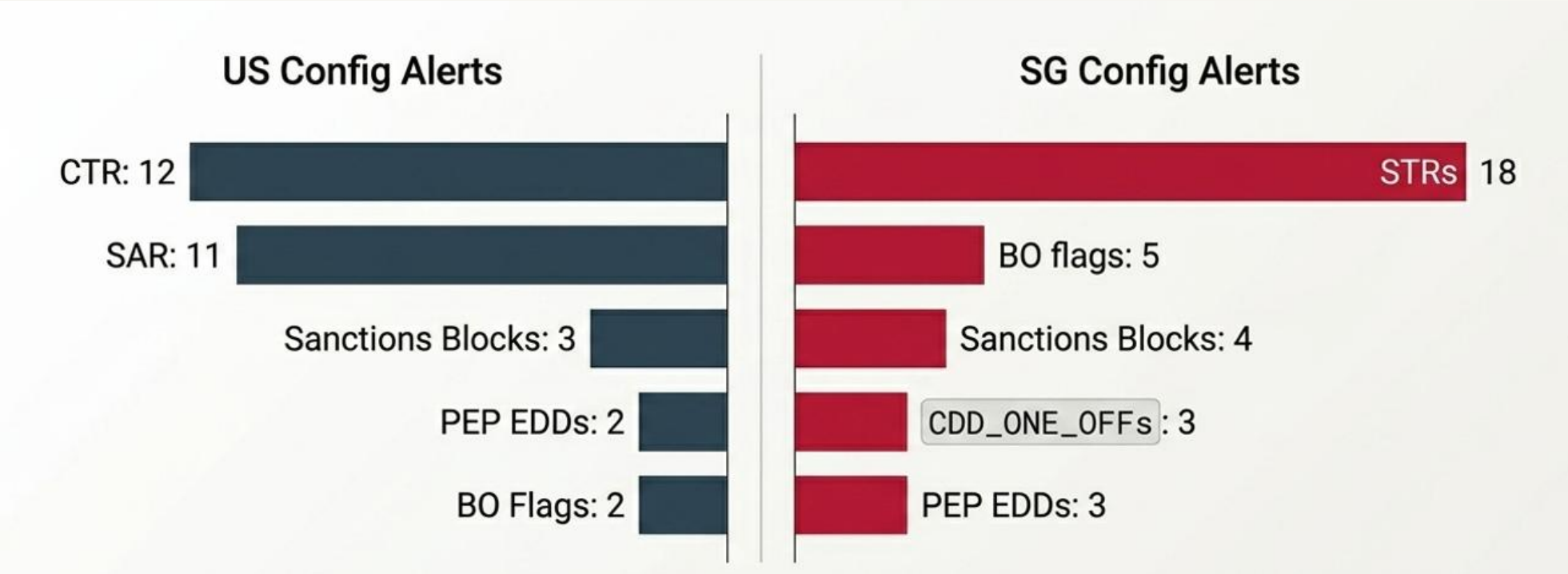
Scenario	US Outcome	SG Outcome
Case A: SGD 18,000 one-off cash exchange	CTR filed (crosses USD 10k equivalent)	Silent (below SGD 20k CDD limit)
Case B: SGD 3,000 trade finance to Myanmar	Silent (below USD 5k SAR floor)	STR filed (high-risk country, no floor)
Case C: SGD 8,000 withdrawal by domestic PEP	Silent (no domestic PEP rule)	EDD_PEP fired (domestic PEP rules)
Case D: Corporate trade finance, 24.9% undisclosed UBO	Silent (escapes 25% bright-line)	STR + BO_FLAG (fails reasonable measures)

The Blind Spot: A Worked Example (TXN-SG-007)



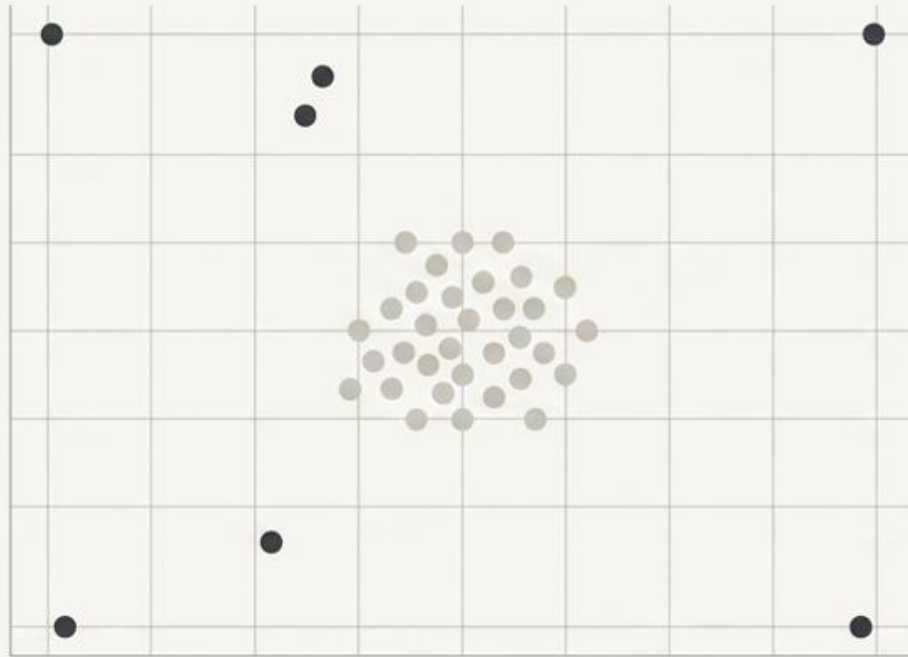
A US-calibrated global engine systematically misses this Singapore legal obligation.

Prototype Results: Complete Regulatory Coverage

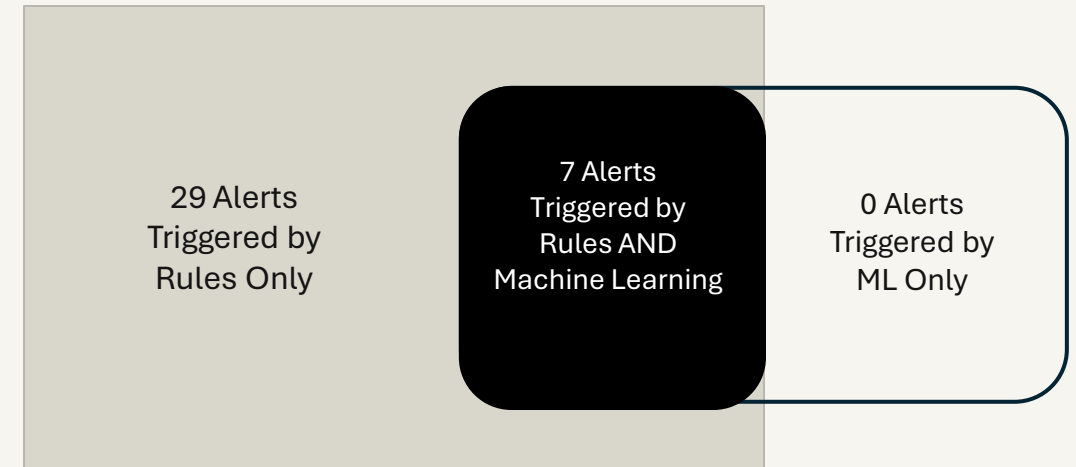


Structuring detector: 7 US smurfing attempts and 6 SG smurfing attempts isolated.

Machine Learning as an Advisory Validator

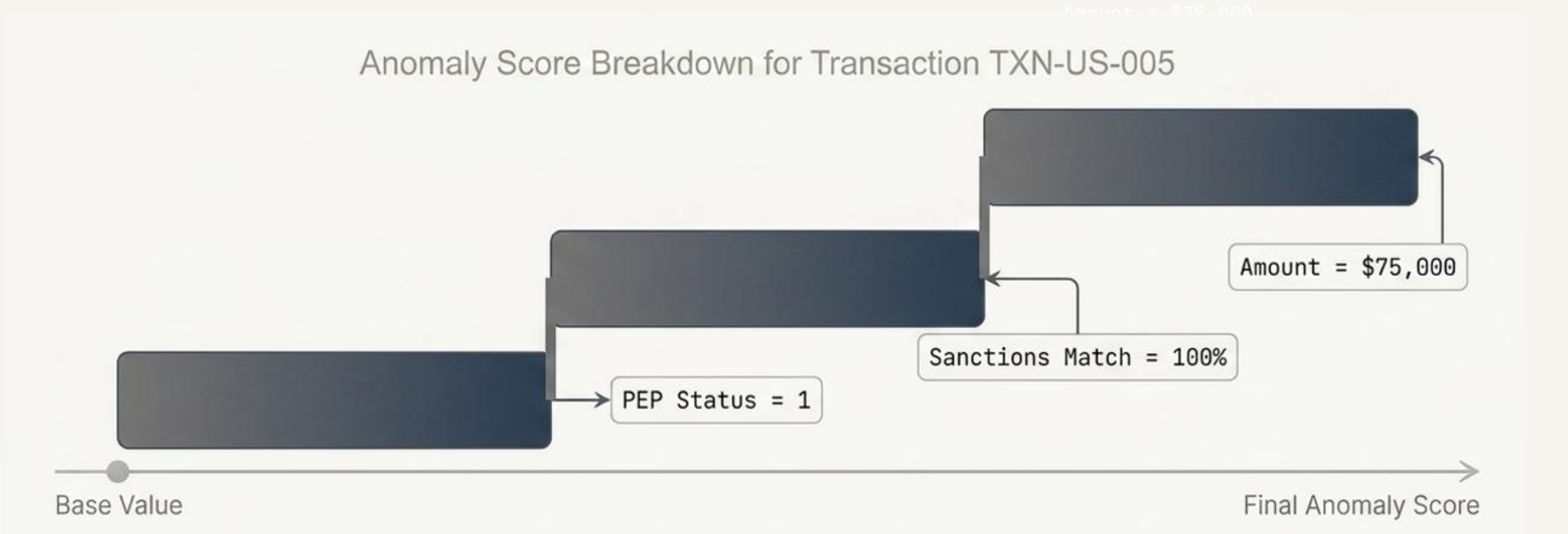


Anomalies Identified (Contamination = 0.15)



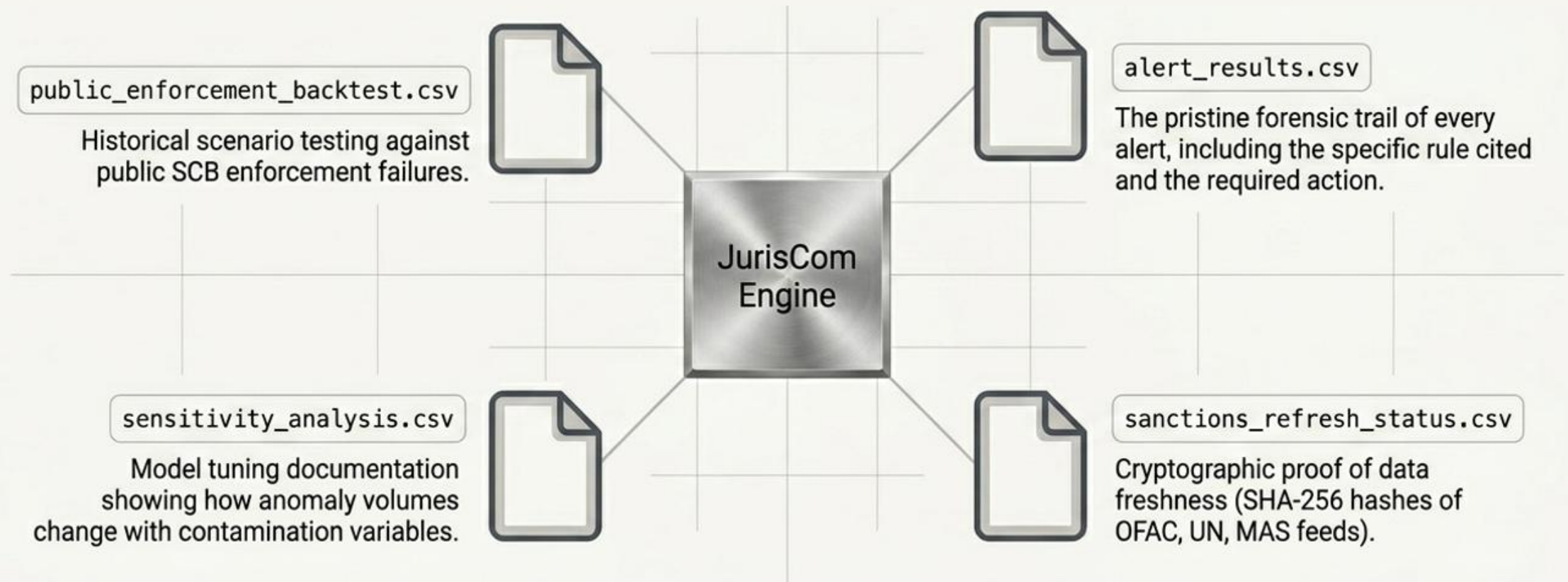
The ML layer flagged 7 of 42 transactions as anomalous. It acts as a validator and prioritizer, never as a standalone legal decision engine.

Opening the AI Black Box: SHAP Explainability



Model scores are legally useless without explanations. SHAP (Shapley Additive exPlanations) reveals why Isolation Forest flagged a transaction, allowing human analysts to cite specific, data-driven reasons in SAR/STR filings.

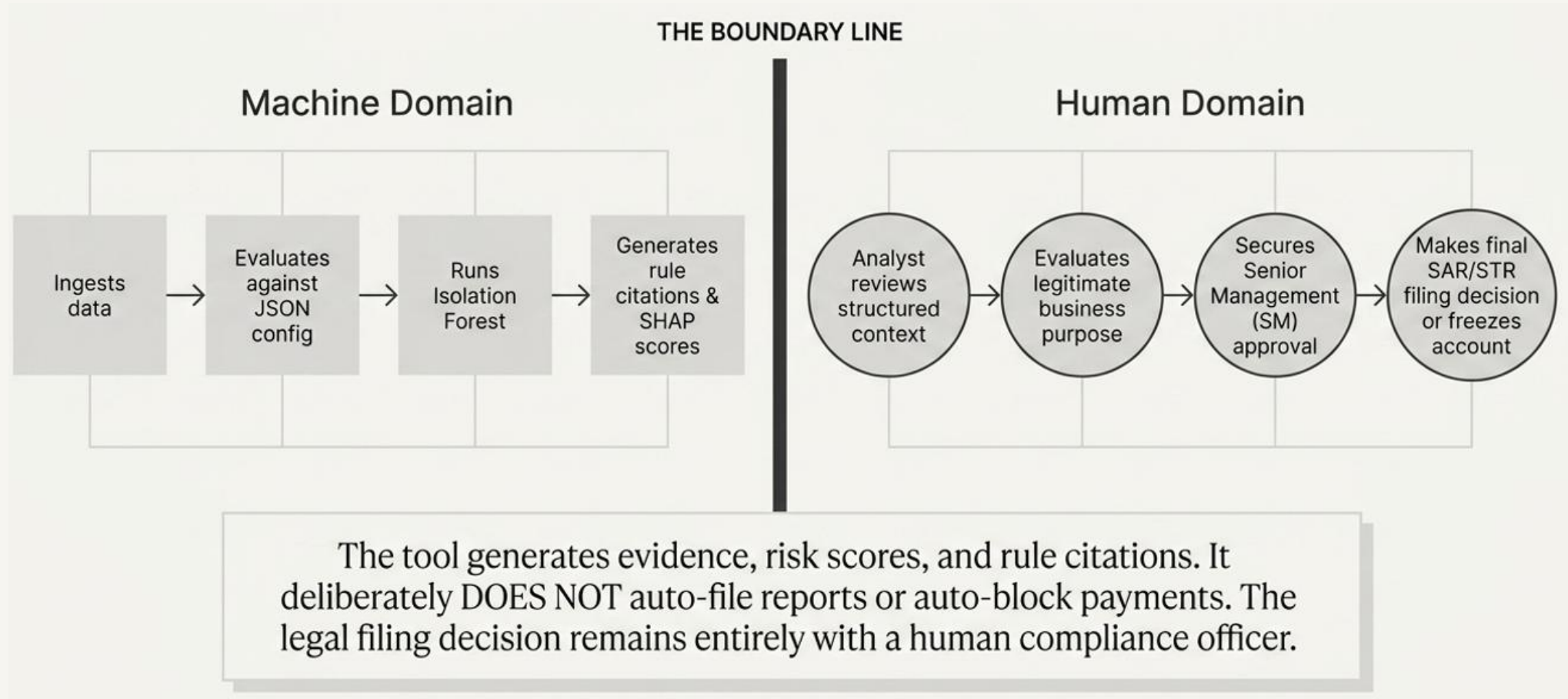
The Evidence Ecosystem: Out-of-the-Box Auditability



The JSON version history and generated artifacts provide a defensible audit trail for examination.

Judgement Cannot Be Outsourced

Focus Question: :What does this product not do, and why?



Scope and Limitations

Scope (What it does flawlessly) ✓	✗ Out of Scope (What it deliberately does not do)
■ Jurisdiction-aware alert bifurcation	■ Real SCB Data: Uses 42 synthetic edge cases, not live production data.
■ Rolling-window structuring detection	■ SWIFT Tampering: Cannot detect stripped SWIFT messages; assumes inputs are visible.
■ SHAP explainability for anomalies	■ Live MAS Parsing: Relies on human review for deeply nested PDF designations.
■ Audit trail generation	■ Graph Topologies: Does not resolve complex multi-layer corporate ownership chains.

Proactive transparency builds credibility.